MLP Task 2

Team ID: 61

Student Name	Student ID	Section Number	Department
نورهان مدحت محمد عشري	2023170682	Section 6	CS
يمني ايمن محمد بيومي	2023170707	Section 6	CS
امنيه مصطفي جوده محمد	2023170101	Section 1	CS
هنا مازن عبدالمولي محمد	2023170690	Section 6	CS
يوسف محمد عاطف احمد	2023170739	Section 4	SC

Testing Results

Test Number	Activation Function	Layers	Hidden Nodes	Learning Rate	Epochs	Bias	Train Accuracy	Test Accuracy
Test 1	Sigmoid	2	3,4	0.01	1000	Yes	98.89%	100.0%
Test 2	Tanh	1	5	0.001	5000	Yes	100.0%	98.33%
Test 3	Sigmoid	1	4	0.1	500	No	100.0%	98.33%
Test 4	Tanh	2	4,4	0.05	1000	Yes	100.0%	100.0%
Test 5	Sigmoid	1	2	0.001	200	Yes	65.56%	66.67%

Test 1 (Sigmoid, 2 hidden layers: 3, 4):

Conclusion: The Sigmoid activation successfully classified the test set with 100% accuracy, demonstrating a classic, gradual convergence curve over 1000 epochs due to its smaller gradient updates.

Test 2 (Tanh, 1 hidden layer: 5):

Conclusion: Using Tanh on Z-score scaled data resulted in instant memorization of the training set (100% accuracy) and highly stable generalization (98.33% test accuracy) across 5000 epochs.

Test 3 (Fast Sigmoid, No Bias):

Conclusion*:* By removing the bias and increasing the learning rate to 0.1, the network converged much faster (500 epochs), proving that the 5 selected features are distinct enough to be linearly separated without a bias shift.

Test 4 (Deep Tanh, 2 hidden layers: 4, 4):

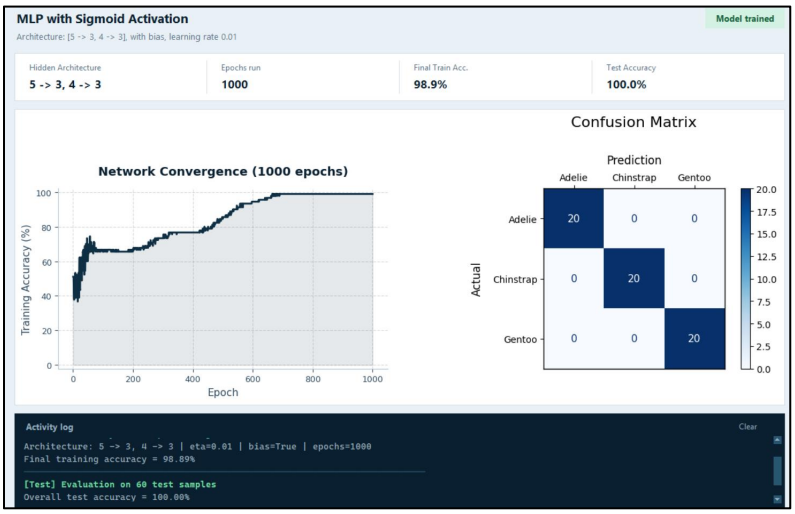
Conclusion: Tanh handled the deeper two-layer architecture flawlessly, avoiding the vanishing gradient problem entirely to achieve a perfect 100% accuracy on both training and unseen test data.

Test 5 (Underfitting Example):

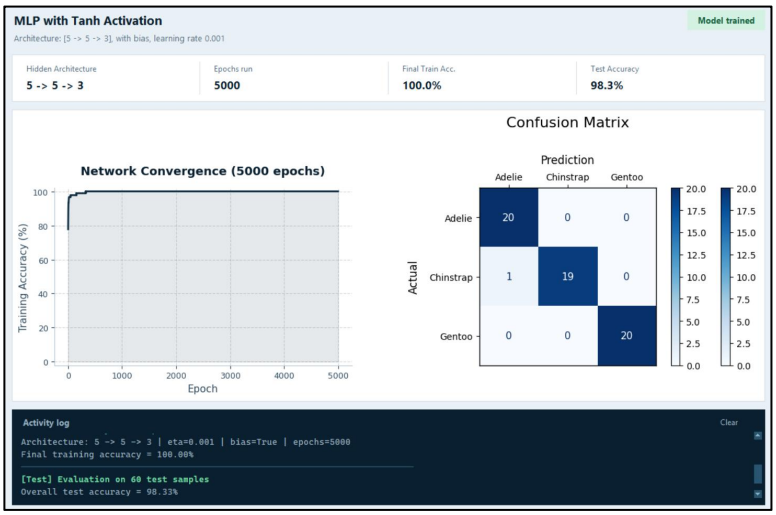
Conclusion: Restricting the architecture to only 2 hidden nodes alongside a tiny learning rate (0.001) and low epoch count (200) resulted in severe underfitting (66.67% accuracy), as the network lacked the internal capacity and time to learn the complex decision boundaries.

Visualizations

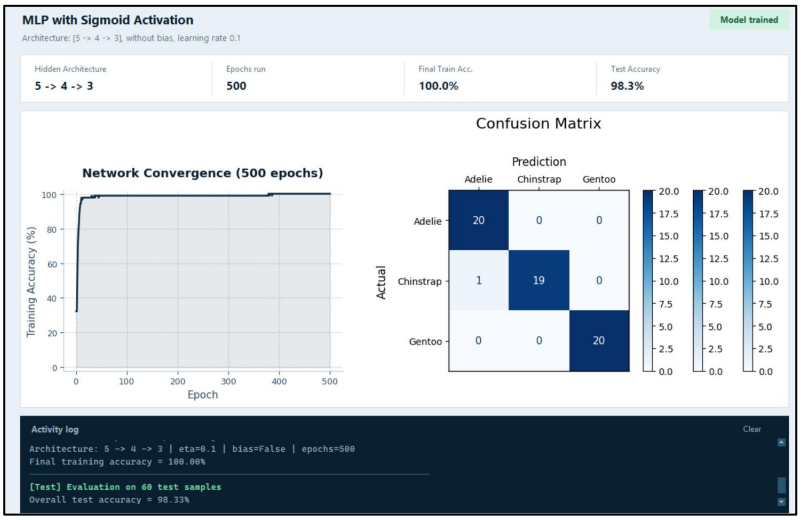
Test 1



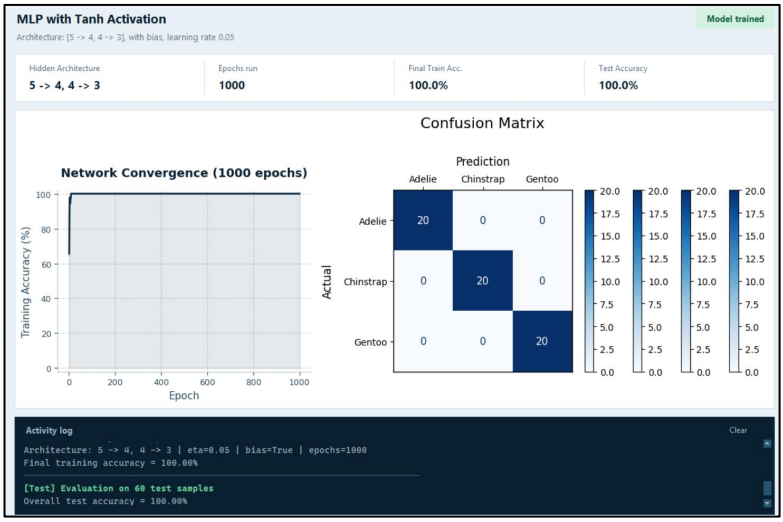
Test 2



Test 3



Test 4



Test 5

